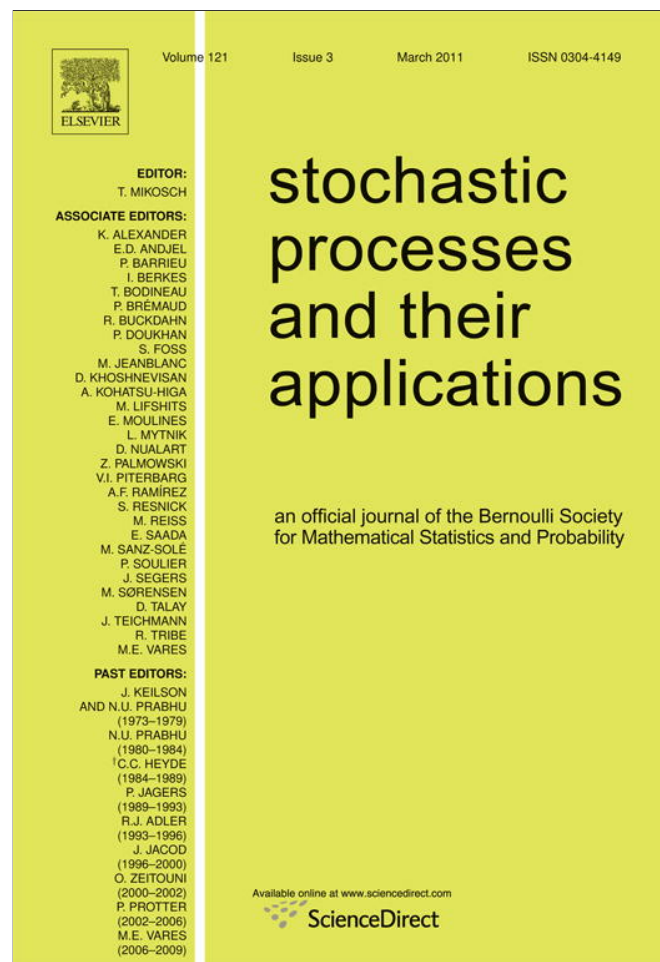




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One-dimensional BSDEs with finite and infinite time horizons[☆]

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Abstract

This paper is devoted to solving one-dimensional backward stochastic differential equations (BSDEs), where the time horizon may be finite or infinite and the assumptions on the generator g are not necessary to be uniform on t . We first show the existence of the minimal solution for this kind of BSDEs with linear growth generators. Then, we establish a general comparison theorem for solutions of this kind of BSDEs with weakly monotonic and uniformly continuous generators. Finally, we give an existence and uniqueness result for solutions of this kind of BSDEs with uniformly continuous generators.

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1. Introduction

We consider the following one-dimensional backward stochastic differential equation (BSDE for short in the remaining):

$$y_t = \xi + \int_t^T g(s, y_s, z_s) ds - \int_t^T z_s \cdot dB_s, \quad t \in [0, T], \quad (1)$$

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where $T > 0$ is a finite or an infinite constant called the time horizon, ξ is a square integral random variable called the terminal condition, the random function $g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbf{R} \times \mathbf{R}^d \rightarrow \mathbf{R}$ is progressively measurable for each (y, z) , called the generator of the BSDE (1), and B is a d -dimensional Brownian motion. The solution (y, z) is a pair of square integrable, adapted processes. The triple (ξ, T, g) is called the parameters of the BSDE (1).

Such equations, in the nonlinear case, were first introduced by Pardoux and Peng [16], who established an existence and uniqueness result for solutions to multidimensional BSDEs under the Lipschitz assumption of the generator g . Since then, BSDEs have been studied with great interest, and they have gradually become an important mathematical tool in many fields such as financial mathematics, stochastic games and optimal control, partial differential equations, etc.

Even though there exist some improvements of the Pardoux–Peng result for multidimensional BSDEs (see e.g. [15,17,8,11,1], and in particular, [6] and [7] where the time horizon T may be infinite and the assumption with respect to the generator g is not necessary to be uniform with respect to t), the case of real-valued BSDEs meaning that y is a real process is easier to handle (see e.g. [13,2,4,3] etc.). In particular, provided that $0 < T < +\infty$ and the assumptions with respect to the generator g are uniform with respect to t . Lepeltier and San Martin [14] showed the existence and uniqueness of the minimal solution for BSDEs when their generators g are continuous and have a linear growth, and Jia [12] studied the existence and uniqueness of the solution for BSDEs when their generator g are independent of y and uniformly continuous in z . The main reason for the gap between the real case and the multidimensional one comes from the comparison theorem. Roughly speaking, a comparison theorem for BSDEs allows one to compare solutions to BSDEs by comparing the terminal conditions and the generators. Since the pioneering work by Peng [18] on the comparison principle for BSDEs (see also Theorem 2.2 in [9]), three general comparison theorems were respectively obtained by Cao and Yan [5], Chen and Wang [7] and Briand and Hu [2]. By the way, let us also mention that, in our knowledge, the monotonicity condition and the uniform continuity condition of the generator g in y were first put forward in [17,11] respectively.

Motivated by these results, we are interested in solving one-dimensional BSDEs, where the time horizon T may be finite or infinite and the assumptions with respect to the generator g are not necessary to be uniform with respect to t . The paper is organized as follows. Section 2 is devoted to the existence of the minimal solution for this kind of BSDEs when their generators g are continuous and have a linear growth (see Theorem 1 in Section 2), which improves the results in [14]. Furthermore, in Section 3, we establish a general comparison theorem for solutions of this kind of BSDEs with generators satisfying some kind of weak monotonicity condition in y and uniform continuity condition in z (see Theorem 2 in Section 3), which can be seen as an extension for four classical comparison theorems mentioned above, Theorem 2.2 in [9], Theorem 2.1 in [5], Theorem A.2 in [7] and Proposition 5 in [2]. Finally, in Section 4, by virtue of the above two results we give an existence and uniqueness result for solutions of this kind of BSDEs with generators satisfying some kind of strong uniform continuity condition in y and uniform continuity condition in z (see Theorem 3 in Section 4). This generalizes some corresponding results in [16,15,7,8,11,12] in the one-dimensional setting.

Let us close this introduction by giving the notations, which we will use in this paper. For what follows, let us fix a number $0 < T \leq +\infty$. Let $(\Omega, \mathcal{F}, P)$ be a probability space carrying a standard d -dimensional Brownian motion $(B_t)_{t \geq 0}$, let $(\mathcal{F}_t)_{t \geq 0}$ be the natural σ -algebra generated by $(B_t)_{t \geq 0}$ and assume $\mathcal{F}_T = \mathcal{F}$. For every positive integer n , we use $|\cdot|$ to denote the norm of a Euclidean space $\mathbf{R}^n$. For $t \in [0, T]$, let $L^2(\Omega, \mathcal{F}_t, P)$ denote the set of all $\mathcal{F}_t$ -measurable random variables ξ such that $E[|\xi|^2] < +\infty$. Let $\mathcal{S}^2(0, T; \mathbf{R})$ be the set of real-valued, adapted

and continuous processes $(Y_t)_{t \in [0, T]}$ such that

$$\|Y\|_{\mathcal{S}^2}^2 := \mathbf{E} \left[\sup_{t \in [0, T]} |Y_t|^2 \right] < +\infty.$$

Moreover, for each positive integer n , let $M^2(0, T; \mathbf{R}^n)$ denote the set of $(\mathcal{F}_t)$ -progressively measurable $\mathbf{R}^n$ -valued processes $\{Z_t, t \in [0, T]\}$ such that

$$\|Z\|_{M^2}^2 \triangleq \mathbf{E} \left[\int_0^T |Z_t|^2 dt \right] < +\infty.$$

Finally, let $\mathbf{S}$ be the set of all nondecreasing continuous functions $\rho(\cdot)$ from $\mathbf{R}^+$ to itself with $\rho(0) = 0$ and $\rho(u) > 0$ for $u > 0$.

As mentioned above, we will deal only with the real-valued BSDE which is an equation of type (1), where the terminal condition $\xi \in L^2(\Omega, \mathcal{F}_T, P)$ and the generator g is $(\mathcal{F}_t)$ -progressively measurable for each (y, z) . In this paper, by a solution to the BSDE (1) we mean a pair of processes (y, z) in $\mathcal{S}^2(0, T; \mathbf{R}) \times M^2(0, T; \mathbf{R}^d)$ which satisfies the BSDE (1) and y is a continuous process.

2. The existence of the minimal solution

In this section, we will show the existence of a minimal solution for BSDEs when their generators g are continuous and have a linear growth (see Theorem 1), which improves the results in [14]. First, we introduce the following assumptions with respect to the generator of the BSDE where $0 < T \leq +\infty$.

(H1) g has a linear growth, i.e., there exist two deterministic functions $u(\cdot), v(\cdot) : [0, T] \mapsto \mathbf{R}^+$ with $\int_0^T [u(t) + v^2(t)] dt < +\infty$, and an $(\mathcal{F}_t)$ -progressively measurable, nonnegative process $\{f_t\}_{t \in [0, T]}$ with $\mathbf{E} \left[\left(\int_0^T f_t dt \right)^2 \right] < +\infty$ such that

$$dP \times dt - \text{a.s.}, \quad \forall y, z, \quad |g(\omega, t, y, z)| \leq f_t(\omega) + u(t)|y| + v(t)|z|.$$

(H2) g is continuous, i.e., $dP \times dt - \text{a.s.}, g(\omega, t, \cdot, \cdot) : \mathbf{R} \times \mathbf{R}^d \mapsto \mathbf{R}$ is a continuous function. The main result of this section is the following Theorem 1.

Theorem 1. Assume that $0 < T \leq +\infty$ and that g satisfies (H1) and (H2). Then for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, the BSDE with parameters (ξ, T, g) has a minimal solution $(y_u, z_u)_{u \in [0, T]}$, which means that if $(\hat{y}_u, \hat{z}_u)_{u \in [0, T]}$ is any solution to the BSDE with parameters (ξ, T, g) , then for each $t \in [0, T]$, $y_t \leq \hat{y}_t, dP - \text{a.s.}$

Remark 1. In the case of $0 < T < +\infty$, it is clear that Theorem 1 generalizes the corresponding result in [14] by taking $u(t) = v(t) = f_t = K$, ($K > 0$). When $T = +\infty$, it is known that Lepeltier's existence result does not work, so Theorem 1 improves his result.

The main idea of the proof of Theorem 1 is motivated by Lepeltier and San Martin [14]. First, we introduce Lemmas 1 and 2, and Proposition 1. The following assumptions will be used, where we assume that $0 < T \leq +\infty$.

(A1) There exist two deterministic functions $u(\cdot), v(\cdot) : [0, T] \mapsto \mathbf{R}^+$ with $\int_0^T [u(t) + v^2(t)] dt < +\infty$ such that $dP \times dt - \text{a.s.},$

$$\forall y_1, y_2, z_1, z_2, \quad |g(\omega, t, y_1, z_1) - g(\omega, t, y_2, z_2)| \leq u(t)|y_1 - y_2| + v(t)|z_1 - z_2|.$$

$$(A2) \mathbf{E} \left[\left(\int_0^T |g(t, 0, 0)| dt \right)^2 \right] < +\infty.$$

Lemma 1 (See Theorem 1.2 in [7]). Assume that $0 < T \leq +\infty$ and g satisfies (A1) and (A2). Then for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, the BSDE with parameters (ξ, T, g) has a unique solution.

Lemma 2 (See Theorem A.2 in [7]). Assume that $0 < T \leq +\infty$ and that g_1 or g_2 satisfies (A1) and (A2). Let $(y_t^1, z_t^1)_{t \in [0, T]}$ and $(y_t^2, z_t^2)_{t \in [0, T]}$ be, respectively, the solution to the BSDE with parameters (ξ_1, T, g_1) and (ξ_2, T, g_2) . If $dP \times dt - a.s.$, $g_1(t, y, z) \leq g_2(t, y, z)$ for each $(y, z) \in \mathbf{R} \times \mathbf{R}^d$ and $dP - a.s.$, $\xi_1 \leq \xi_2$, then, we have, for each $t \in [0, T]$, $dP - a.s.$, $y_t^1 \leq y_t^2$.

Using the similar proof of Lemma 1 in [14], one can obtain the following proposition. We omit its proof.

Proposition 1. Assume that the generator g satisfies (H1) and (H2). Let g_n be the function defined as follows:

$$g_n(\omega, t, y, z) := \inf_{(u, v) \in \mathbf{R}^{1+d}} \{g(\omega, t, u, v) + nu(t)|y - u| + nv(t)|z - v|\}.$$

Then the sequence of functions g_n is well defined for each $n \geq 1$, and it satisfies, $dP \times dt - a.s.$

- (i) Linear growth: $\forall y, z, |g_n(\omega, t, y, z)| \leq f_t(\omega) + u(t)|y| + v(t)|z|$.
- (ii) Monotonicity in n : $\forall y, z, g_n(\omega, t, y, z)$ increases in n .
- (iii) Lipschitz condition: $\forall y_1, y_2, z_1, z_2$, we have

$$|g_n(\omega, t, y_1, z_1) - g_n(\omega, t, y_2, z_2)| \leq nu(t)|y_1 - y_2| + nv(t)|z_1 - z_2|.$$

- (iv) Convergence: If $(y_n, z_n) \rightarrow (y, z)$, then $g_n(\omega, t, y_n, z_n) \rightarrow g(\omega, t, y, z)$.

Now, we can begin the proof of Theorem 1.

Sketch of Proof of Theorem 1. Let g_n be defined as in Proposition 1. In view of (iii) and (i) of Proposition 1 together with (H1), Lemma 1 shows that, for each $n \geq 1$, the BSDEs with parameters (ξ, T, g_n) and (ξ, T, h) have a unique solution $(y_u^n, z_u^n)_{u \in [0, T]}$ and $(Y_u, Z_u)_{u \in [0, T]}$ respectively, where $h(\omega, t, y, z) := f_t(\omega) + u(t)|y| + v(t)|z|$, $\forall (\omega, t, y, z)$. And, in view of (i) and (ii) of Proposition 1, the comparison theorem (Lemma 2) yields that for each $n \geq 1$, $y_t^1(\omega) \leq y_t^n(\omega) \leq y_t^{n+1}(\omega) \leq Y_t(\omega)$, $dP \times dt - a.s.$ Thus, there must exist an $(\mathcal{F}_t)$ -progressively measurable process $(y_u)_{u \in [0, T]}$ satisfying

$$\lim_{n \rightarrow \infty} y_t^n(\omega) = y_t(\omega), \quad dP \times dt - a.s. \quad (2)$$

Now, let $G(\omega) := \sup_n \sup_{s \in [0, T]} |y_s^n(\omega)|$. We have

$$\mathbf{E} \left[\sup_{s \in [0, T]} |y_s|^2 \right] \leq \mathbf{E} [G^2] < +\infty. \quad (3)$$

Applying Itô's formula to $(y_t^n)^2$ and using (i) of Proposition 1 with (H1), from the inequalities $2ab \leq a^2 + b^2$, $2ab \leq a^2\lambda + b^2/\lambda < +\infty$ where $\lambda := 2 \int_0^T v^2(s)ds$ and Hölder's inequality we can deduce that

$$\mathbf{E} \left[\int_0^T |z_s^n|^2 ds \right] \leq \mathbf{E} |\xi|^2 + 2\mathbf{E} \left[\int_0^T |y_s^n| (f_s + u(s)|y_s^n| + v(s)|z_s^n|) ds \right]$$

$$\begin{aligned} &\leq \mathbf{E}|\xi|^2 + (1 + \lambda)\mathbf{E}[G^2] + \mathbf{E}\left[\left(\int_0^T f_s ds\right)^2\right] \\ &\quad + 2\mathbf{E}[G^2] \cdot \int_0^T u(s)ds + \frac{1}{\lambda}\mathbf{E}\left[\left(\int_0^T v(s)|z_s^n|ds\right)^2\right] \\ &= C_1 + \frac{1}{2}\mathbf{E}\left[\int_0^T |z_s^n|^2 ds\right], \end{aligned}$$

where the constant C_1 is independent of n , and then $\sup_n \mathbf{E}\left[\int_0^T |z_s^n|^2 ds\right] \leq 2C_1 < +\infty$.

Furthermore, applying Itô's formula to $(y_t^n - y_t^m)^2$ and using (i) in Proposition 1 and Hölder's inequality one has

$$\begin{aligned} &\mathbf{E}|y_0^n - y_0^m|^2 + \mathbf{E}\left[\int_0^T |z_s^n - z_s^m|^2 du\right] \\ &\leq 4\mathbf{E}\left[\int_0^T |y_s^n - y_s^m|f_s ds\right] + 4\sqrt{\mathbf{E}[G^2]} \cdot \sqrt{\mathbf{E}\left[\left(\int_0^T |y_s^n - y_s^m|u(s)ds\right)^2\right]} \\ &\quad + 2\sqrt{8C_1}\sqrt{\mathbf{E}\left[\int_0^T |y_s^n - y_s^m|^2 v^2(s)ds\right]}. \end{aligned}$$

Thus, in view of (3) and (H1), Lebesgue's dominated convergence theorem yields that (z^n) is a Cauchy sequence in $M^2(0, T; \mathbf{R}^d)$, then (z^n) converges in $M^2(0, T; \mathbf{R}^d)$. We shall denote by z the limit of (z^n) , that is to say,

$$\lim_{n \rightarrow \infty} \mathbf{E} \int_0^T |z_s^n - z_s|^2 ds = 0. \quad (4)$$

Furthermore, we can assume, passing to a subsequence if necessary, that, $dP \times dt$ - a.s., $z^n \rightarrow z$. and $\mathbf{E} \int_0^T \sup_n |z_s^n|^2 ds < +\infty$. Therefore, in view of (iv) of Proposition 1 and (2) we get that as $n \rightarrow \infty$,

$$g_n(s, y_s^n, z_s^n) \rightarrow g(s, y_s, z_s), \quad dP \times dt - \text{a.s.}$$

On the other hand, it follows from (H1) and (i) of Proposition 1 that for each $n \geq 1$, $dP \times dt$ - a.s.,

$$|g_n(s, y_s^n, z_s^n) - g(s, y_s, z_s)| \leq 2 \left(f_s + u(s)G + v(s) \sup_n |z_s^n| \right).$$

Thus, Lebesgue's dominated convergence theorem yields that

$$\lim_{n \rightarrow \infty} \mathbf{E} \left[\left(\int_0^T |g_n(s, y_s^n, z_s^n) - g(s, y_s, z_s)| ds \right)^2 \right] = 0. \quad (5)$$

At the same time, we also have $dP \times dt$ - a.s., for each $n, m \geq 1$ and $t \in [0, T]$,

$$\begin{aligned} \mathbf{E} \left[\sup_{t \in [0, T]} |y_t^n - y_t^m|^2 \right] &\leq 2\mathbf{E} \left[\left(\int_0^T |g_n(s, y_s^n, z_s^n) - g_m(s, y_s^m, z_s^m)| ds \right)^2 \right] \\ &\quad + 16\mathbf{E} \left[\int_0^T |z_s^n - z_s^m|^2 ds \right], \end{aligned}$$

where we have used BGD's inequality. Thus, in view of (2), from the above inequality, (4) and (5) we can assume that

$$\lim_{n \rightarrow \infty} \mathbf{E} \left[\sup_{t \in [0, T]} |y_t^n - y_t|^2 \right] = 0.$$

from which one knows that $y_t \in \mathcal{S}^2(0, T; \mathbf{R})$. Thus, in view of the above identity, (4) and (5), by taking limits in BSDEs with parameters (ξ, T, g_n) we deduce that $(y_t, z_t)_{t \in [0, T]}$ is a solution of the BSDE with parameters (ξ, T, g) .

Finally, let $(\hat{y}_t, \hat{z}_t)_{t \in [0, T]}$ be any solution of the BSDE with parameters (ξ, T, g) , by the comparison theorem (Lemma 2) and in view of (ii) and (iii) of Proposition 1, we have dP -a.s., $y_t^n \leq \hat{y}_t$ for each $t \in [0, T]$ and $n \in \mathbf{N}$, then $y_t \leq \hat{y}_t$ proving that $(y_t, z_t)_{t \in [0, T]}$ is the minimal solution of the BSDE with parameters (ξ, T, g) . The proof of Theorem 1 is complete. $\square$

The following two examples show that under the assumptions of Theorem 1, the solution of the BSDE with parameters (ξ, T, g) is not unique in general.

Example 1. Let $0 < T < +\infty$, $\xi = 0$ and $g(t, y, z) = \sqrt{\frac{|y|}{t}}$ for each $(y, z) \in \mathbf{R}^{1+d}$. One can directly verify that for each $t_0 \in [0, T]$, $(y_t^{t_0}, 0)_{t \in [0, T]}$ are all the solutions of the BSDE with parameters (ξ, T, g) , where

$$y_t^{t_0} = \begin{cases} \left(\sqrt{t_0} - \sqrt{t} \right)^2, & \text{for } 0 \leq t < t_0, \\ 0, & \text{for } t_0 \leq t \leq T. \end{cases}$$

Example 2. Let $0 < T \leq +\infty$, $\xi = 0$ and $g(t, y, z) = \frac{\sqrt[4]{|y|}}{(1+t)^2}$ for each $(y, z) \in \mathbf{R}^{1+d}$. One can directly verify that for each $t_0 \in [0, T]$, $(y_t^{t_0}, 0)_{t \in [0, T]}$ are all the solutions of the BSDE with parameters (ξ, T, g) , where

$$y_t^{t_0} = \begin{cases} \frac{3}{4} \left(\frac{1}{1+t} - \frac{1}{1+t_0} \right)^{4/3}, & \text{for } 0 \leq t < t_0, \\ 0, & \text{for } t_0 \leq t \leq T. \end{cases}$$

3. The general comparison theorem

In this section, we will establish a general comparison theorem for solutions of BSDEs with generators satisfying some kind of weak monotonicity condition in y and uniform continuity condition in z (see Theorem 2), which can be seen as an extension for four classical comparison theorems, Theorem 2.2 in [9], Theorem 2.1 in [5], Theorem A.2 in [7] (see Lemma 2 in Section 2) and Proposition 5 in [2]. Before that, let us introduce three technical lemmas. First, the following Lemma 3 gives a sequence of upper bounds for functions of linear growth, which comes from Lemma 4 in [10].

Lemma 3. Let $\psi(\cdot) : \mathbf{R}^+ \mapsto \mathbf{R}^+$ be a nondecreasing function of linear growth, which means that $\psi(x) \leq K(x + 1)$ ($K > 0$) holds true for all $x \in \mathbf{R}^+$. Then for each $n \geq 2K$,

$$\psi(x) \leq nx + \psi\left(\frac{2K}{n}\right)$$

holds true for each $x \in \mathbf{R}^+$.

Second, similar to the proof of Gronwall's inequality, one can obtain the following [Lemma 4](#). We omit its proof.

Lemma 4. Let $a > 0$, $0 \leq t \leq T \leq +\infty$ and $\beta(r) : [0, T] \mapsto \mathbf{R}^+$ satisfy $\int_0^T \beta(s)ds < +\infty$. Let $u(r) : [t, T] \mapsto \mathbf{R}^+$ be a continuous function with $\sup_{r \in [t, T]} u(r) < +\infty$ and satisfying

$$u(r) \leq a + \int_r^T \beta(s)u(s)ds, \quad r \in [t, T].$$

Then

$$u(r) \leq ae^{\int_r^T \beta(s)ds}, \quad r \in [t, T].$$

Finally, by virtue of [Lemma 4](#), we can deduce the following [Lemma 5](#), which plays an important role in the proof of [Theorem 2](#).

Lemma 5. Let $0 \leq t \leq T \leq +\infty$, $\beta(t) : [0, T] \mapsto \mathbf{R}^+$ satisfy $\int_0^T \beta(s)ds < +\infty$ and let $\{u_n(r), r \in [t, T]\}_{n=1}^{+\infty}$ be a sequence of nonnegative and continuous functions with $\sup_{r \in [t, T]} u_n(r) < +\infty$ such that

$$u_n(r) \leq b_n + \int_r^T \beta(s)\psi(u_n(s))ds, \quad r \in [t, T], \quad (6)$$

where $\{b_n\}_{n=1}^{+\infty}$ is a nonnegative real sequence with $\lim_{n \rightarrow \infty} b_n = 0$, and $\psi(\cdot)$ belongs to $\mathbf{S}$, increases at most linearly and satisfies $\int_{0+} \frac{1}{\psi(u)} du = +\infty$. Then, we have

$$\forall n \geq 1, \quad u_n(r) \leq \left(\sup_{n \geq 1} b_n + K \int_0^T \beta(s)ds \right) \cdot e^{K \int_0^T \beta(s)ds}, \quad r \in [t, T]. \quad (7)$$

where K is the constant of the linear growth for $\psi(\cdot)$, i.e., $\forall u \geq 0$, $\psi(u) \leq K(1+u)$. And, for each $r \in [t, T]$, the limit of the sequence $\{u_n(r)\}_{n=1}^{+\infty}$ is zero.

Proof. Since $\forall u \geq 0$, $\psi(u) \leq K(1+u)$, in view of $\sup_{n \geq 1} b_n < +\infty$ and $\int_0^T \beta(s)ds < +\infty$, from (6) one knows that for each $n \geq 1$,

$$u_n(r) \leq \sup_{n \geq 1} b_n + K \int_t^T \beta(s)ds + \int_r^T K\beta(s)u_n(s)ds, \quad r \in [t, T].$$

Then, in view of $\sup_{r \in [t, T]} u_n(r) < +\infty$, (7) follows from [Lemma 4](#) immediately.

In what follows, it follows from (7) that for each $r \in [t, T]$, $f(r) := \overline{\lim_{n \rightarrow \infty}} u_n(r) < +\infty$ and $\sup_{r \in [t, T]} f(r) < \infty$. Moreover, in view of the fact that $\psi(\cdot)$ is a nondecreasing continuous function, from Fatou's convergence theorem we can deduce that

$$\overline{\lim_{n \rightarrow \infty}} \int_r^T \beta(s)\psi(u_n(s))ds \leq \int_r^T \beta(s)\psi\left(\overline{\lim_{n \rightarrow \infty}} u_n(s)\right)ds, \quad r \in [t, T]. \quad (8)$$

By taking limits in (6), in view of $\lim_{n \rightarrow \infty} b_n = 0$ and (8), one gets

$$f(r) \leq \int_r^T \beta(s)\psi(f(s))ds, \quad r \in [t, T]. \quad (9)$$

Since $\psi(\cdot)$ increases at most linearly and $\int_0^T \beta(s) ds < +\infty$, one can deduce that the function

$$R(r) = \int_r^T \beta(s) \psi(f(s)) ds, \quad r \in [t, T]$$

takes its values in $\mathbf{R}^+$, $R(T) = 0$ and $R'(r) = -\beta(r) \psi(f(r))$, $dt - a.e.$, $r \in [t, T]$. Thus, recalling that $\psi(\cdot)$ is a nondecreasing function, from (9) we have

$$\frac{-R'(s)}{\psi(R(s))} = \frac{\beta(s) \psi(f(s))}{\psi(R(s))} \leq \beta(s), \quad dt - a.e., \quad s \in [t, T].$$

and then for each $r \in [t, T]$,

$$\int_0^{f(r)} \frac{1}{\psi(u)} du \leq \int_{R(T)}^{R(r)} \frac{1}{\psi(u)} du = \int_r^T \frac{-R'(s)}{\psi(R(s))} ds \leq \int_r^T \beta(s) ds < +\infty.$$

Thus, since $\int_{0+} \frac{1}{\psi(u)} du = +\infty$, one knows that for each $r \in [t, T]$, $f(r)$ must be zero. Then we complete the proof of Lemma 5. $\square$

Now, we can turn to the expression of our main result in this section, Theorem 2. Let us further introduce the following assumptions on the generator g of the BSDE (1), where we assume that $0 < T \leq +\infty$.

(H3) g is weakly monotonic in y , i.e., there exists a deterministic function $u(\cdot) : [0, T] \mapsto \mathbf{R}^+$ with $\int_0^T u(t) dt < +\infty$ and a concave function $\rho(\cdot) \in \mathbf{S}$ with $\int_{0+} \frac{1}{\rho(u)} du = +\infty$ such that $dP \times dt - a.s.$,

$$\forall y_1, y_2, z, \quad \text{sgn}(y_1 - y_2) \cdot (g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \leq u(t) \rho(|y_1 - y_2|).$$

(H4) g is uniformly continuous in z uniformly with respect to (ω, y) , i.e., there exists a deterministic function $v(\cdot) : [0, T] \mapsto \mathbf{R}^+$ with $\int_0^T v(t) dt < +\infty$ and a function $\phi(\cdot) \in \mathbf{S}$ of linear growth such that $dP \times dt - a.s.$,

$$\forall y, z_1, z_2, |g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq v(t) \phi(|z_1 - z_2|).$$

Here and henceforth, we always assume that $0 \leq \phi(x) \leq ax + b$ for all $x \in \mathbf{R}^+$. Furthermore, we also assume that $\int_0^T v(t) dt < +\infty$ when $b \neq 0$.

Remark 2. The inequality in (H3) may be written as follows: $dP \times dt - a.s.$,

$$\forall y_1, y_2, z, \quad (y_1 - y_2) \cdot (g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \leq |y_1 - y_2| u(t) \rho(|y_1 - y_2|).$$

It is clear that (H3) is weaker than the monotonicity condition in y put forward by Pardoux [17]. Furthermore, the function $\rho(\cdot)$ in (H3) increases at most linearly since it is a nondecreasing concave function and $\rho(0) = 0$. Here and henceforth, we denote the constant of linear growth for $\rho(\cdot)$ by k , i.e., $0 \leq \rho(x) \leq k(x + 1)$ for all $x \in \mathbf{R}^+$.

Remark 3. If $0 < T < +\infty$, then it follows from Hölder's inequality that the condition $\int_0^T v(t) dt < +\infty$ in (H4) holds true automatically.

The following Theorem 2 establishes a general comparison theorem for BSDEs under assumptions (H3) and (H4), which is the main result in this section.

Theorem 2. Let $0 < T \leq +\infty$, $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, g and g' be two generators of BSDEs, and let $(y_t, z_t)_{t \in [0, T]}$ and $(y'_t, z'_t)_{t \in [0, T]}$ be, respectively, a solution to the BSDE with parameters

(ξ, T, g) and (ξ', T, g') . If $dP - a.s.$, $\xi \leq \xi'$, g (resp. g') satisfies (H3) and (H4) and $dP \times dt - a.s.$, $g(t, y'_t, z'_t) \leq g'(t, y'_t, z'_t)$ (resp. $g(t, y_t, z_t) \leq g'(t, y_t, z_t)$), then for each $t \in [0, T]$, we have

$$dP - a.s., \quad y_t \leq y'_t.$$

Proof. We first assume that $dP - a.s.$, $\xi \leq \xi'$, g satisfies (H3) and (H4) and $dP \times dt - a.s.$, $g(t, y'_t, z'_t) \leq g'(t, y'_t, z'_t)$. Tanaka's formula leads to the equation, setting $\hat{y}_t = y_t - y'_t$, $\hat{z}_t = z_t - z'_t$,

$$\hat{y}_t^+ \leq (\xi - \xi')^+ + \int_t^T 1_{\hat{y}_s > 0} [g(s, y_s, z_s) - g'(s, y'_s, z'_s)] ds - \int_t^T 1_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s. \quad (10)$$

First of all, since $g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)$ is non-positive, we have

$$\begin{aligned} g(s, y_s, z_s) - g'(s, y'_s, z'_s) &= g(s, y_s, z_s) - g(s, y'_s, z'_s) + g(s, y'_s, z'_s) - g'(s, y'_s, z'_s) \\ &\leq g(s, y_s, z_s) - g(s, y'_s, z_s) + g(s, y'_s, z_s) - g(s, y'_s, z'_s) \end{aligned}$$

and we deduce, using the weak monotonicity in y and the uniform continuity in z of g , that

$$1_{\hat{y}_s > 0} [g(s, y_s, z_s) - g'(s, y'_s, z'_s)] \leq u(s) \rho(\hat{y}_s^+) + 1_{\hat{y}_s > 0} v(s) \phi(|\hat{z}_s|). \quad (11)$$

In what follows, applying Lemma 3 with $\psi(\cdot) = \phi(\cdot)$ and $K = c := a + b$, we have

$$\forall n \geq 2c, x \in \mathbf{R}^+, \quad \phi(x) \leq nx + 1_{b \neq 0} \phi\left(\frac{2c}{n}\right). \quad (12)$$

Thus, considering $\xi \leq \xi'$ we can get from (10)–(12) that, for each $n \geq 2c$ and each $t \in [0, T]$,

$$\begin{aligned} \hat{y}_t^+ &\leq a_n + \int_t^T [u(s) \rho(\hat{y}_s^+) + 1_{\hat{y}_s > 0} n v(s) |\hat{z}_s|] ds - \int_t^T 1_{\hat{y}_s > 0} \hat{z}_s \cdot dB_s \\ &= a_n + \int_t^T u(s) \rho(\hat{y}_s^+) ds - \int_t^T 1_{\hat{y}_s > 0} \hat{z}_s \cdot \left[-\frac{n v(s) \hat{z}_s}{|\hat{z}_s|} 1_{|\hat{z}_s| \neq 0} ds + dB_s \right], \end{aligned} \quad (13)$$

where

$$a_n = 1_{b \neq 0} \phi\left(\frac{2c}{n}\right) \cdot \int_0^T v(s) ds \rightarrow 0 \quad \text{by (H4) as } n \rightarrow \infty.$$

Let P_n be the probability on $(\Omega, \mathcal{F})$ which is equivalent to P and defined by:

$$\frac{dP_n}{dP} := \exp \left\{ n \int_0^T \frac{v(s) \hat{z}_s}{|\hat{z}_s|} 1_{|\hat{z}_s| \neq 0} \cdot dB_s - \frac{1}{2} n^2 \int_0^T 1_{|\hat{z}_s| \neq 0} v^2(s) ds \right\}.$$

It is worth noting that dP_n/dP has moments of all order since $\int_0^T v^2(s) ds < +\infty$ and $v(\cdot)$ is a deterministic function. By Girsanov's theorem, under P_n the process

$$B_n(t) = B_t - \int_0^t \frac{n \hat{z}_s}{|\hat{z}_s|} 1_{|\hat{z}_s| \neq 0} ds, \quad t \in [0, T],$$

is an $(\mathcal{F}_t, P_n)$ -Brownian motion. Moreover, the process $\left(\int_0^t 1_{\hat{y}_s > 0} \hat{z}_s \cdot dB_n(s) \right)_{0 \leq t \leq T}$ is an $(\mathcal{F}_t, P_n)$ -martingale. Indeed, let $\mathbf{E}_n[X|\mathcal{F}_t]$ represent the conditional expectation of the random variable X with respect to $\mathcal{F}_t$ under P_n and let $\mathbf{E}_n[X] \triangleq \mathbf{E}_n[X|\mathcal{F}_0]$, then from BGD's inequality

and Hölder's inequality, we have

$$\begin{aligned} \mathbf{E}_n \left[\sup_{0 \leq t \leq T} \left| \int_0^t 1_{\hat{y}_s > 0} \hat{z}_s \cdot dB_n(s) \right| \right] &\leq C \mathbf{E}_n \left[\sqrt{\int_0^T |\hat{z}_s|^2 ds} \right] \\ &\leq C \sqrt{\mathbf{E} \left[\left(\frac{dP_n}{dP} \right)^2 \right]} \sqrt{\mathbf{E} \left[\int_0^T |\hat{z}_s|^2 ds \right]} < +\infty, \end{aligned}$$

where C is a constant. Thus, by taking the conditional expectation with respect to $\mathcal{F}_t$ under P_n in (13) and noticing that $u(\cdot)$ and $v(\cdot)$ are deterministic and $\rho(\cdot)$ is concave, from Fubini's theorem and Jensen's inequality one gets that for each $n \geq 2c$ and $t \in [0, T]$,

$$\mathbf{E}_n [\hat{y}_r^+ | \mathcal{F}_t] \leq a_n + \int_r^T u(s) \rho(\mathbf{E}_n [\hat{y}_s^+ | \mathcal{F}_t]) ds, \quad r \in [t, T].$$

Thus, in view of (H3) and Remark 2, applying Lemma 5 with $u_n(r) = \mathbf{E}_n [\hat{y}_r^+ | \mathcal{F}_t]$, $b_n = a_n$, $\beta(s) = u(s)$, $\psi(u) = \rho(u)$ and $K = k$ we can get that $\hat{y}_t^+ = \lim_{n \rightarrow \infty} u_n(t) = 0$. That is to say, for each $t \in [0, T]$, dP - a.s., $y_t \leq y'_t$.

Now, let us assume that dP - a.s., $\xi \leq \xi'$, g' satisfies (H3) and (H4) and $dP \times dt$ - a.s., $g(t, y_t, z_t) \leq g'(t, y_t, z_t)$. Then, since $g(s, y_s, z_s) - g'(s, y_s, z_s)$ is non-positive, we have

$$\begin{aligned} g(s, y_s, z_s) - g'(s, y'_s, z'_s) &= g(s, y_s, z_s) - g'(s, y_s, z_s) + g'(s, y_s, z_s) - g'(s, y'_s, z'_s) \\ &\leq g'(s, y_s, z_s) - g'(s, y'_s, z_s) + g'(s, y'_s, z_s) - g'(s, y'_s, z'_s) \end{aligned}$$

and using the weak monotonicity in y and the uniform continuity in z of g' , one knows that the inequality (11) holds still true. Therefore, the same proof as above yields that for each $t \in [0, T]$, dP - a.s., $y_t \leq y'_t$. Theorem 2 is proved. $\square$

Remark 4. Theorem 2 shows that the comparison theorem for BSDEs with their generators g satisfying (H3) and (H4) holds true. However, the associated strict comparison theorem does not hold in general, readers can find a counter example in [12]. The main reason is assumption (H4).

From Theorem 2, the following two corollaries are immediate.

Corollary 1. Let $0 < T \leq +\infty$, and let both g and g' satisfy assumptions (H3) and (H4). Let $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, and let $(y_t, z_t)_{t \in [0, T]}$ and $(y'_t, z'_t)_{t \in [0, T]}$ be, respectively, a solution to the BSDE with parameters (ξ, T, g) and (ξ', T, g') . If dP - a.s., $\xi \leq \xi'$ and $dP \times dt$ - a.s., $\forall y, z$, $g(t, y, z) \leq g'(t, y, z)$, then for each $t \in [0, T]$, dP - a.s., $y_t \leq y'_t$.

Corollary 2. Assume that $0 < T \leq +\infty$, g satisfy (H3) and (H4), and $\xi \in L^2(\Omega, \mathcal{F}_T, P)$. Then the BSDE with parameters (ξ, T, g) has at most one solution.

Example 3. Let $0 < T \leq +\infty$ and $g(t, y, z) = e^{-y} + \frac{1}{(1+t)^2} (q(|y|) + \sqrt{|z|})$, where

$$q(x) = \begin{cases} x \ln \frac{1}{x}, & 0 < x \leq \delta; \\ q'(\delta-)(x - \delta) + q(\delta), & x > \delta; \\ 0, & \text{other cases} \end{cases}$$

with $\delta > 0$ small enough.

Obviously, g satisfies (H4) by choosing $\phi(x) = \sqrt{x}$ with $v(t) = \frac{1}{(1+t)^2}$. One can also prove that g satisfies (H3) with $\rho(x) = q(x)$ and $u(t) = \frac{1}{(1+t)^2}$ by verifying that $\int_{0+} \frac{du}{q(u)} = +\infty$ and $q(\cdot)$ is nondecreasing and concave, and the following inequality holds true

$$\forall y_1 > y_2, z, \quad g(\omega, t, y_1, z) - g(\omega, t, y_2, z) \leq \frac{1}{(1+t)^2} q(|y_1 - y_2|).$$

Then, from Theorem 2 or Corollary 2 one knows that for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, the BSDE with parameters (ξ, T, g) has at most one solution.

It is worth mentioning that this g does not satisfy assumption (H1). Nevertheless we can prove that there exists a solution to the BSDE with parameters (ξ, T, g) and then the solution is unique by Theorem 2, which will be discussed in another paper by us.

Remark 5. It is clear that Theorem 2 and Corollary 1 generalizes the comparison theorem in [7] (see Lemma 2) by taking $\rho(x) = \phi(x) = x$.

In order to compare Theorem 2 and Corollary 1 with some another classical comparison theorems, let us further introduce the following assumptions which are stronger than (H3) and (H4), where we assume that $0 < T < +\infty$.

(H3') There exists a concave function $\rho(\cdot) \in \mathbf{S}$ with $\int_{0+} \frac{1}{\rho(u)} du = +\infty$ such that, $dP \times dt - a.s.$,

$$\forall y_1, y_2, z, \quad \text{sgn}(y_1 - y_2) \cdot (g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \leq \rho(|y_1 - y_2|).$$

(H3'a) g is monotonic in y , i.e., there exists a constant $\mu \geq 0$, such that, $dP \times dt - a.s.$,

$$\forall y_1, y_2, z, \quad (y_1 - y_2) \cdot (g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \leq \mu |y_1 - y_2|^2.$$

(H3'b) There exists a concave function $\rho(\cdot) \in \mathbf{S}$ with $\int_{0+} \frac{1}{\rho(u)} du = +\infty$ such that, $dP \times dt - a.s.$,

$$\forall y_1, y_2, z, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)| \leq \rho(|y_1 - y_2|).$$

(H3'b') g satisfies the Mao's non-Lipschitz condition in y , i.e., there exists a concave function $\rho(\cdot) \in \mathbf{S}$ with $\int_{0+} \frac{1}{\rho(u)} du = +\infty$ such that, $dP \times dt - a.s.$,

$$\forall y_1, y_2, z, \quad |g(\omega, t, y_1, z) - g(\omega, t, y_2, z)|^2 \leq \rho(|y_1 - y_2|^2).$$

(H4') There exists a function $\phi(\cdot) \in \mathbf{S}$ of linear growth such that, $dP \times dt - a.s.$,

$$\forall y, z_1, z_2, \quad |g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq \phi(|z_1 - z_2|).$$

(H4'') g is Lipschitz continuous in z uniformly with respect to (ω, t, y) , i.e., there exists a constant $C \geq 0$ such that, $dP \times dt - a.s.$,

$$\forall y, z_1, z_2, \quad |g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq C |z_1 - z_2|.$$

Remark 6. Theorem 2.1 in [5] and Proposition 5 in [2] respectively established a comparison theorem for BSDEs under assumptions (H3'b') and (H4''), and assumptions (H3'a) and (H4''), which are both a generalization for Theorem 2.2 in [9]. Theorem 2 and Corollary 1 also extend these three classical comparison theorems since $(H4'') \implies (H4') \implies (H4)$, $(H3'a) \implies (H3')$ $\implies (H3)$, and $(H3'b') \implies (H3'b) \implies (H3')$. It is worth mentioning that none of the inverse versions of the implication relations mentioned above holds true.

4. The existence and uniqueness result

As applications of [Theorems 1 and 2](#), we will establish a general uniqueness and existence theorem (see [Theorem 3](#)) for BSDEs in this section, which generalizes the corresponding results in [\[16,15,7,8,11,12\]](#) in the one-dimensional setting. We need the following assumption on g , where we also assume that $0 < T \leq +\infty$.

(H5) g is strongly uniformly continuous in y and uniformly continuous in z , i.e., $dP \times dt$ -a.s., for each y_1, y_2, z_1, z_2 ,

$$|g(\omega, t, y_1, z_1) - g(\omega, t, y_2, z_2)| \leq u(t)\rho(|y_1 - y_2|) + v(t)\phi(|z_1 - z_2|),$$

where $u(\cdot)$ and $\rho(\cdot)$ are defined in (H3), and $v(\cdot)$ and $\phi(\cdot)$ are defined in (H4).

Remark 7. By the classical theory of uniformly continuous functions, the concavity condition with respect to $\rho(\cdot)$ in (H3'b), (H3'b') and (H5) can be taken away. Furthermore, the condition of linear growth with respect to $\phi(\cdot)$ in (H4), (H4') and (H5) can also be eliminated.

The following [Theorem 3](#) is the main result of this section.

Theorem 3. Assume that $0 < T \leq +\infty$ and that g satisfies (H5) and (A2) in [Section 2](#). Then for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, the BSDE with parameters (ξ, T, g) has a unique solution.

Proof. The uniqueness result follows from [Theorem 2](#) or [Corollary 2](#) due to the fact that (H5) implies (H3) and (H4). In what follows, we show the existence part. In fact, it follows from assumption (H5) that

$$\begin{aligned} |g(\omega, t, y, z)| &\leq |g(\omega, t, y, z) - g(\omega, t, 0, 0)| + |g(\omega, t, 0, 0)| \\ &\leq u(t)\rho(|y|) + v(t)\phi(|z|) + |g(\omega, t, 0, 0)| \\ &\leq ku(t)|y| + av(t)|z| + |g(\omega, t, 0, 0)| + ku(t) + bv(t) \end{aligned}$$

Thus, in view of (A2) and (H5), we know that g satisfies assumptions (H2) and (H1) with $f_t = |g(\omega, t, 0, 0)| + ku(t) + bv(t)$. Thus, the existence result follows from [Theorem 1](#), and the proof is complete. $\square$

Remark 8. It is clear that [Theorem 3](#) generalizes the corresponding result in [\[7\]](#) (see [Lemma 1](#) in [Section 2](#)) by taking $\rho(x) = \phi(x) = x$.

In particular, by letting $0 < T < +\infty$ and $u(t) = v(t) = 1$ in assumption (H5) of [Theorem 3](#) we obtain the following corollary.

Corollary 3. Assume that $0 < T < +\infty$ and g satisfies assumptions (H3'b), (H4') and (A2). Then for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, the BSDE with parameters (ξ, T, g) has a unique solution.

Remark 9. In view of [Remark 6](#) and Hölder's inequality, it is easy to see that [Theorem 3](#) and [Corollary 3](#) are also an improvement for the corresponding (one-dimensional) ones in [\[15,11,12,16,8\]](#).

Finally, according to [Theorem 3](#) we introduce three examples of BSDEs which have a unique solution.

Example 4. Let $0 < T < +\infty$ and

$$g(t, y, z) = \frac{1}{\sqrt{t}} q(|y|) + \frac{1}{\sqrt[4]{t}} |z|^\alpha + |B_t|,$$

where $\alpha \in [0, 1]$ and $q(\cdot)$ is defined in Example 3.

Obviously, g satisfies (A2). Similar to Example 3, it is not difficult to prove that g also satisfies (H5) with $u(t) = \frac{1}{\sqrt{t}}$, $\rho(x) = q(x)$, $v(t) = \frac{1}{\sqrt[4]{t}}$ and $\phi(x) = x^\alpha$. Thus, from Theorem 3 one knows that the BSDE with parameters (ξ, T, g) has a unique solution for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$.

Example 5. Let $0 < T \leq +\infty$ and

$$g(t, y, z) = \frac{1}{(1+t)^2} q(|y|) + \frac{1}{1+t} \sin |z|,$$

where $q(\cdot)$ is also defined in Example 3.

Similar to Example 4, it is not difficult to prove that g also satisfies (A2) and (H5) with $u(t) = \frac{1}{(1+t)^2}$, $\rho(x) = q(x)$, $v(t) = \frac{1}{1+t}$ and $\phi(x) = x$ with $a = 1$ and $b = 0$. Thus, from Theorem 3 one knows that the BSDE with parameters (ξ, T, g) has a unique solution for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$.

Example 6. Let $0 < T \leq +\infty$ and

$$g(t, y, z) = \frac{1}{1+t^2} q(|y|) + \frac{1}{(1+t)^2} \sqrt{|z|} + \frac{1}{(1+t)^2},$$

where $q(\cdot)$ is also defined in Example 3.

Similar to Example 4, it is not difficult to prove that g also satisfies (A2) and (H5) with $u(t) = \frac{1}{1+t^2}$, $\rho(x) = q(x)$, $v(t) = \frac{1}{(1+t)^2}$ and $\phi(x) = \sqrt{x}$. Thus, from Theorem 3 one knows that the BSDE with parameters (ξ, T, g) has a unique solution for each $\xi \in L^2(\Omega, \mathcal{F}_T, P)$.

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